
Mahout Brain 2.0

Planning Handoff for Life Map, Memory Atoms, Care Trail, Vulnerable Windows, and North Star Action Layer

Core product sentence

Brain 2.0 helps Mahout remember, map, and respond to the user's recurring life patterns: what issues appear, when they appear, what Mahout did, what the user did next, and whether the pattern is resolving, returning, or needs care.

Prepared from the June 28, 2026 Brain 2.0 planning discussion.

Working mode: Kotlin + XML only, MVVM, preservation-first, paid/credit-heavy intelligence layer only.

1. Executive summary

Brain 2.0 is not just a better memory page. It is the premium intelligence layer that turns user history into visible, actionable support. It should help paid users see their recurring issues, vulnerable windows, replacement behaviors, and how Mahout responded over time.

Locked direction

Brain 2.0 must be amazing, visible, and not a free feature. Free users may keep operational basics and possibly see locked previews. Paid users or users with credits get the real Life Map, Care Trail, recurring pattern detection, returning issue detection, conversation starters, and stronger North Star action planning.

The five core pillars

- 1 **Memory Atoms 2.0:** more precise issue, trigger, replacement, outcome, time-window, confidence, and safety atoms.
- 2 **Life Map:** a visible premium map of when issues appear across days and time blocks.
- 3 **Care Trail:** what Mahout did about the issue, what the user did next, and whether it helped.
- 4 **Returning Issue Detection:** careful early warning when a previously quiet issue starts showing up again.
- 5 **North Star Action Layer:** morning goal suggestions, conversation starters, and easy save/edit flows inside North Star.

Today's major additions

- **Vulnerable windows** are the time windows where negative or unwanted patterns usually start, rise, or repeat.
- **Conversation starters** are different from normal sparks: they can open a meaningful North Star dialogue at the right time.
- **Daily goals** should be suggested by North Star every morning based on history, aims, patterns, and current state.
- **Path should not be rebuilt now.** Instead, add a beautiful visual guide that teaches time-based coloring and check-off completion.
- **AI auto-linking** should connect Path actions to Aim/future goals when confidence is high, with undo/review.

2. Locked naming and language

The names matter because they shape whether the feature feels clinical, generic, or premium. The following names should be treated as product vocabulary unless a later design pass intentionally changes them.

Term	Meaning	User-facing behavior
Brain 2.0	The paid intelligence upgrade to memory, pattern detection, Life Map, and North Star action support.	Used in planning, release notes, and internal work. In UI, users may see Life Map, Brain, and North Star intelligence surfaces.
Life Map	The visible premium screen showing issues, vulnerable windows, replacements, and care outcomes.	First-class screen under North Star/Brain, not hidden in settings.
Vulnerable Window	A recurring time window where an unwanted issue tends to start or become stronger.	Used in letters and notifications carefully: 'This may be a vulnerable window.'
Care Trail	Timeline of issue -> Mahout response -> user response -> outcome over the next hours/days.	Shown inside issue detail; proves Mahout is following up.
Returning Issue	A previously quiet/resolved/dormant issue that starts appearing again.	Flagged quickly and gently: 'This old pattern may be starting to return.'
Conversation Starter	A proactive North Star prompt designed to begin dialogue at meaningful moments.	More personal than a reminder; can open a paid/credit-using conversation.
Action Layer	North Star's ability to propose, edit, link, and save daily goals/actions.	Morning plan suggestions and quick add/edit/save inside North Star.

Tone rule

Do not say 'failed', 'relapsed', 'cured', 'treated', or 'diagnosed'. Say 'returning pattern', 'high-care pattern', 'catch it early', 'interrupt the loop', 'needs support', and 'what seemed to help'.

3. Premium / paid-only boundary

Brain 2.0 is for the paid side of Mahout. It can also be available during a trial/Bond Pass or through credits, but it should not become a normal free feature. This protects both product value and AI cost.

Surface	Free behavior	Paid / credit behavior
Basic mood, journal, Path, Aim	Operational core remains usable.	Same operational core plus intelligence on top.
Life Map	Locked preview or teaser only; no full map.	Visible 7-day map, evidence sheets, Care Trail, returning issue detection.
Issue pattern detection	Not exposed as a full intelligence layer.	Recurring issues, vulnerable windows, frequency, days/week, time blocks.
Care Trail	No full timeline of Mahout support.	Shows what Mahout did, what user did, and whether it helped.
Conversation starters	Basic reminders only if existing.	Pattern-aware, context-aware, potentially credit-using North Star starts.
Morning daily goals	Basic manual goal/action creation remains.	North Star suggests today's goals from Brain, history, Aim, and Life Map.
AI auto-linking	Maybe manual linking only.	High-confidence auto-linking with undo/review.

Implementation rule

Do not add a separate random feature flag for Brain 2.0. Use the existing monetization policy/resolver system so Free, Pro, Pro Light, Bond Pass, and credit states stay consistent.

4. Memory Atoms 2.0

Memory Atoms 2.0 are the foundation. Life Map cannot feel intelligent unless the Brain stores the right small pieces: issue, trigger, time, replacement, outcome, confidence, and evidence.

Atom categories

Atom category	Examples	Why it matters
Issue atom	Loneliness, alcohol craving, sadness, low energy, anger, procrastination, doom scrolling.	Tells Mahout what pattern is appearing.
Trigger atom	After work, night, argument, skipped sleep, scrolling, Sunday evening, money stress.	Helps Mahout understand why or when the issue starts.
Time-window atom	Morning, afternoon, evening, night, exact time if available.	Creates vulnerable windows and Life Map cells.
Replacement atom	Walk, journal, call friend, read book, Path action, calm reset, workout.	Shows healthier behavior that may interrupt the loop.
Outcome atom	Eased, unchanged, returned, improved mood, ignored, replaced by action.	Tells whether Mahout's support had a positive effect.
Safety atom	Normal, sensitive, high-care.	Controls wording, escalation, and trust boundaries.
Evidence atom	Mood log ID, journal ID, Path action ID, North Star message ID, notification event.	Lets the user tap 'why this?' and trust the map.

Required atom fields

- Stable atom id.
- Type: issue, trigger, replacement, outcome, pattern, preference, life event, safety note.
- Title and short human-readable summary.
- Source: user_said, inferred, telemetry, AI_summary, NorthStar_chat, journal, mood_log, Path.
- Confidence: low, medium, high, or numeric 0.0-1.0 if the repo already uses numeric confidence.
- Confirmation lane: candidate, confirmed, rejected, superseded.
- Sensitivity: normal, sensitive, high_care.
- First seen, last seen, and evidence pointers.
- Visibility: active, hidden, deleted. Hidden/deleted atoms must not enter AI context.
- Superseding pointer for precision upgrades.

Trust rule

Facts from user logs can be stated clearly. Derived patterns must be phrased softly. Inferences from text should be confirmed before becoming durable truth.

5. Life Map visible product design

Life Map must be visible. It cannot only exist inside hidden Brain summaries or North Star prompts. The user should be able to open it and see the map of their week, their vulnerable windows, their issue history, and how Mahout responded.

Best placement

- Primary: North Star -> Brain -> Life Map.
- Optional premium card: North Star header or Brain header: 'Life Map'.
- Do not bury Life Map in Settings.
- Free users see a premium locked preview, not the full intelligence surface.

First version layout

Area	Content	Notes
Header	Life Map + one-line explanation.	Example: 'See what keeps returning, what helps, and where North Star is supporting you.'
Weekly summary	Top 2-4 patterns: returning issue, strongest replacement, unresolved issue, positive shift.	Should be readable before the grid.
7-day map	Columns Mon-Sun. Rows Morning, Afternoon, Evening, Night.	Each cell shows issue chips, replacement chips, or empty calm state.
Cell detail	Exact issues, source, intensity, Mahout response, user response, outcome.	Tap bottom sheet or detail screen.
Issue detail	Pattern state, vulnerable window, frequency, Care Trail, returning comparison.	Most important drill-down.
Controls	Filter by issue, hide/delete issue, exclude from letters/sparks.	Trust and privacy controls are not optional.

Life Map cell examples

Cell	Chips shown	Meaning
Monday evening	Alcohol craving; Walk helped	Issue appeared, replacement happened after.
Tuesday night	Loneliness; Returning	Old issue has started appearing again.
Wednesday morning	Low energy	Issue logged once; maybe open state only.
Thursday evening	Reading; Blessed	Positive replacement or opposite pattern is emerging.
Friday night	No issue; protected window	Vulnerable window passed without issue or with positive action.

6. Vulnerable Windows

A Vulnerable Window is the product language for the time when an unwanted pattern tends to begin, rise, or repeat. This is one of the strongest ideas from today's discussion and should become core to Brain 2.0.

Definition

Vulnerable Window

A recurring time block where a negative or unwanted issue is more likely to show up. It can be based on user logs, mood logs, journal extraction, North Star conversations, Path inactivity, or repeated issue events.

Window phases

Phase	Timing	Mahout behavior
Pre-window	30-60 minutes before the usual issue window.	Prepare gently: suggest replacement action, morning/evening letter note, soft conversation starter.
Active window	During the expected issue time.	Ask how the user is doing, offer comfort/progress choice, suggest one tiny interruption.
Post-window	1-3 hours after the issue or expected issue.	Check outcome: did it ease, return, get replaced, or stay open?
Next-days watch	2-7 days after repeated issue.	Detect resolving, dormant, returning, or needs-care state.

Examples

- Alcohol craving often appears at 8-10pm. Mahout checks in around the beginning of that window and offers a 3-minute reset, walk, journal, or support contact suggestion.
- Loneliness often appears at night. Mahout starts a conversation: 'Do you want comfort, connection, or one tiny action?'
- Low energy repeats in the afternoon. Mahout suggests a lower-friction goal instead of pushing a hard deep work block.
- Sadness after work used to repeat, but reading at night and logging 'blessed' is replacing it. Mahout celebrates and protects the new pattern.

7. Care Trail and positive effect tracking

Care Trail is the proof layer. It shows whether Mahout actually responded to the user's issue and whether anything positive happened afterward. Without Care Trail, Life Map is just analytics. With Care Trail, it becomes evidence of support.

Care Trail answers four questions

- 1 What issue happened and when?
- 2 What did Mahout do about it?
- 3 What did the user do after Mahout responded?
- 4 Did the pattern ease, repeat, return, get replaced, or remain unresolved?

Care Trail timeline structure

Stage	Examples	Data source
Issue detected	8:05pm alcohol craving; intensity 7/10.	IssueEvent, MoodLog, chat extraction, journal extraction.
Mahout response	Spark sent; North Star asked a question; evening letter referenced the window.	Notification event, NorthStar message, Letter entity.
User response	Opened North Star; journaled; started walk; ignored; logged better mood.	Telemetry, journal, Path session, MoodLog.
Outcome	Calmer logged; issue returned next day; no follow-up; positive replacement formed.	OutcomeEvent, later IssueEvents, Path, mood.
Pattern update	Repeating, resolving, dormant, returning, or needs care.	LifeMapPattern engine.

Positive effect language

Avoid	Use instead
Mahout treated your craving.	Mahout helped you interrupt the loop once.
Mahout cured this pattern.	This pattern seemed to ease after the replacement action.
You relapsed.	This old pattern may be returning.
You failed again.	Catching it early is the win.
This proves the cause.	This is a possible signal based on your logs.

Example Care Trail summary

Monday 8:05pm: alcohol craving logged. Mahout sent a check-in and suggested a walk. User started a 12-minute walk at 8:20pm and later logged calmer. Tuesday evening had no craving logged. Wednesday 8:30pm: craving returned. Current read: walking helped once; evening remains a vulnerable window.

8. Returning Issue detection

A Returning Issue is a high-value Brain 2.0 state. It means an issue that had been quiet, easing, resolving, or dormant starts showing up again. It should be handled quickly, carefully, and without shame.

User-facing message

This old pattern may be starting to return. No shame - catching it early is the win.

Pattern states

State	Meaning	North Star behavior
Open	Issue appeared but not enough pattern evidence yet.	Ask one clarifying question or offer simple support.
Repeating	Issue is showing multiple times in similar windows.	Create vulnerable window and suggest early interruption.
Eased	Issue seemed to reduce after support or action.	Record possible helper and mention carefully.
Interrupted	User did replacement action during the window.	Celebrate the loop interruption.
Resolving	Frequency/intensity is trending down.	Reinforce what is working.
Dormant	Issue has been quiet for a meaningful time.	Do not over-mention; keep as background pattern.
Returning	Dormant/resolving issue appears again.	Quick careful check-in; compare last time vs now.
Needs care	High-care or unresolved recurring issue.	Support, safety wording, real-world support suggestion where appropriate.

Returning issue detail should show

- How long the issue was quiet.
- When it first returned.
- Whether it returned again.
- Whether intensity is lower, same, or higher.
- What helped last time.
- Whether the old replacement has been tried again.
- A quick early support option.

Example

Alcohol craving returning

Evening craving was quiet for 12 days. It appeared again Monday at 8:20pm and Wednesday at 8:05pm. Last time, walking helped once and journaling helped once. Suggested early move: protect the 7:30-9:30pm window before the craving gets loud.

9. Conversation Starters and Sparks

Sparks already exist as a concept in Mahout, but Brain 2.0 needs another layer: conversation starters. A spark can be a short nudge. A conversation starter is an invitation into a real North Star exchange based on Life Map context.

Type	Purpose	Example
Spark	Short nudge or prompt, often notification-style.	This is usually a vulnerable window. Want a 3-minute reset?
Conversation Starter	North Star begins a meaningful check-in or support dialogue.	Evening is the window where this old pattern has returned twice. How are you doing right now?
Letter mention	Morning/evening letter uses the pattern as a planning or reflection anchor.	Tonight we protect the window early instead of waiting until it is loud.
Action prompt	Direct path into daily goals or replacement action.	Want to add a 10-minute walk to tonight's plan?

When conversation starters should trigger

- High-confidence vulnerable window is beginning.
- A returning issue appears after a dormant period.
- A high-care issue repeats and has not been followed by a positive replacement.
- A positive replacement pattern is forming and deserves reinforcement.
- User opened North Star during a known window.
- Morning plan needs to protect a known vulnerable window.

Credit / paid behavior

- The simple notification itself can be deterministic and cheap.
- Opening a generated North Star exchange can spend credits or use Pro entitlement.
- Aggressive credit use is acceptable only when the interaction is high-value and clearly tied to Brain 2.0 evidence.
- Do not create filler conversations just to increase usage.

Do not spam

Conversation starters should feel like a thoughtful friend showing up at the right moment. They should not feel like a nag, streak reminder, or generic push notification.

10. North Star Action Layer and daily goals

The action layer may be just as important as Life Map. The user should not always have to go to Aim/Mountain or Path and manually build a plan. North Star should use history, aims, patterns, and current state to propose useful daily goals every morning.

Action layer product sentence

Every morning, North Star should suggest a small set of daily goals based on what the user is aiming at, what they usually do, what they have been struggling with, and what windows need protection.

Morning daily goals flow

- 1 North Star checks Aim/future goals, previous daily goals, Path history, Life Map, vulnerable windows, and unresolved/returning issues.
- 2 It proposes 3-5 Today Goals or action cards.
- 3 The user can edit, delete, add, reorder, or save inside North Star without jumping to another tab.
- 4 Each proposed goal can link to a Path action or create a lightweight Today Goal.
- 5 Saved goals appear in the right existing place: Today Goals / Path / Aim receipts, depending on current repo structure.
- 6 North Star can follow up in the evening: what was done, what was skipped, what helped.

Suggested goal sources

Source	How it affects suggestions
Active Aim/future goals	Keeps the daily plan aligned with the user's bigger direction.
Past daily goals	Repeats what the user consistently wants to do, without asking again every morning.
Path actions	Suggests concrete actions that already exist and can be started or checked.
Life Map vulnerable windows	Adds protective goals like walk, journal, read, call someone, calm reset.
Returning issues	Prioritizes early interruption, not large ambition.
Positive replacements	Reinforces what is replacing the old issue: reading at night, workouts, gratitude, social connection.
Energy/mood state	Chooses tiny vs normal step; avoids unrealistic goals during low-energy/high-care windows.

User control

- The user should be able to save the plan in one tap.
- The user should be able to edit any card inline.
- Deleting a suggestion should not delete the underlying saved goal/action unless explicitly chosen.
- North Star should not ask too many questions before providing value.
- Default should be useful even if user does nothing but tap Save.

11. Path 2.0 support: visual guide, not a rebuild

Path should be made easier by teaching it better, not by redesigning the interaction at this stage. The existing segment coloring / time action flow should remain stable. The improvement is a premium-quality visual guide that shows how to use Path quickly.

What the visual must teach

- How to add a time-based action.
- How to color/select time segments for a time-based action.
- How to start or log a timer/session if the current UI supports it.
- How to complete/check a check action if check actions exist in the current repo.
- How the logged action feeds goals and North Star receipts.
- How Path connects back to Brain 2.0 through replacements and Care Trail.

Format options

Option	Benefit	Risk
Static illustrated guide	Fast, safe, premium-looking, no behavior risk.	Less dynamic than animation.
Dismissible first-run overlay	Shows at exactly the right time.	Must not annoy returning users.
Short step carousel	Easy to explain 3-4 interactions.	Needs careful layout work.
Inline empty-state card	Useful when user has no Path logs yet.	Less visible after data exists.
Animation/GIF-like sequence	Very clear if done well.	Higher asset and QA risk.

Recommendation

For launch safety, create a static premium step carousel or illustrated guide first. Do not change Path mechanics. Do not rewrite the timeline. Do not touch timer logic unless the current slice explicitly requires it.

12. AI auto-linking between goals and Path actions

Auto-linking reduces friction. Users should not have to manually connect every action to every goal when the relationship is obvious. This is a strong paid feature because it makes the app feel like it is doing the mental work.

Core behavior

Moment	Mahout behavior
New goal created	Mahout checks existing Path actions and suggests or applies high-confidence links.
New Path action created	Mahout checks active goals and suggests or applies high-confidence links.
Morning daily goals	North Star proposes goals/actions already linked when confidence is high.
Weekly review	Mahout can surface unlinked actions that look goal-relevant.
User correction	If user unlinks or rejects, Mahout learns and avoids repeating the same wrong link.

High-confidence auto-link examples

- Goal: Reduce weight by 10 kg. Actions: running 30 minutes, workout 1 hour, skip meal/check nutrition. Link them to the weight goal.
- Goal: Complete Mahout programming in two months. Action: coding. Link coding receipts to Mahout programming goal.
- Goal: Build another app. Action: coding. If the action is generic, link only when the user goal context or action note supports both, or ask/review.
- Goal: Read 12 books. Action: reading at night. Link reading receipts to the reading goal and positive replacement pattern if it replaces sadness/drinking.

Default policy

Best balance

High-confidence links should be applied immediately with a small review/undo message. Medium-confidence links should ask: 'Do you want to link these actions?' Low-confidence links should not be made.

User-facing messages

- New goal: 'I found 3 Path actions that can feed this goal: Running, Workout, and Nutrition. I linked them - you can undo any of them.'
- New action: 'This looks like it supports Reduce weight by 10 kg. I linked it so your time counts toward that goal.'
- Medium confidence: 'Are these actions going to help you achieve this goal?'
- Weekly cleanup: 'Coding is feeding two app goals. Want to keep both links or choose one primary goal?'

13. Practical data model

The data model should be additive and repo-aware. Do not replace current memory tables or Path/Aim models unless Cursor proves the current structure cannot support the slice.

Conceptual models

Model	Purpose	Key fields
BrainAtom2	Small durable memory unit.	id, type, title, source, confidence, confirmation, sensitivity, firstSeen, lastSeen, evidenceRefs, hidden/deleted.
IssueEvent	A moment where an issue appears.	id, issueType, timestamp, timeBlock, intensity, note, source, confidence, moodId, journalId, sensitivity.
MahoutInterventionEvent	What Mahout did in response.	id, issueEventId/patternId, type, timestamp, surface, messageId, notificationId, suggestionType.
OutcomeEvent	What happened after intervention.	id, issueEventId, interventionId, outcomeType, timestamp, evidenceRef, confidence.
LifeMapCell	Display cell for day/time block.	date, timeBlock, issueChips, replacementChips, state, evidenceCount.
LifeMapPattern	Recurring pattern summary.	issueType, windows, daysPerWeek, frequency, lastSeen, trend, state, bestReplacement, confidence.
CareTrail	Detailed timeline for issue/pattern.	patternId, events, interventions, outcomes, nextDays, summary.
DailyGoalSuggestion	North Star proposed daily action.	id, title, reason, sourceSignals, linkedGoalId, linkedActionId, editable, confidence.

Evidence references

Evidence references should point back to existing user data rather than copying everything. A Life Map pattern can reference mood logs, Path actions/sessions, journal entries, North Star messages, letters, and notification interactions.

Deletion rule

If a user hides or deletes an issue, atom, or evidence item, it must disappear from Life Map UI and future AI context. Historical UI should not crash if old references are missing.

14. Pattern engine rules

The first Brain 2.0 pattern engine should be deterministic. AI can enrich language, but the source of truth should be local facts and simple rules.

Core calculations

Calculation	Rule idea
Time block	Map timestamp into Morning, Afternoon, Evening, Night using the app's existing time-block convention or a single new shared utility.
Days per week	Count unique days with issue event in rolling 7-day window.
Recurring issue	Same issue appears 2+ times in same or adjacent time block within 7-14 days.
Vulnerable window	Recurring issue with consistent time block/day pattern and enough confidence.
Dormant issue	Previously recurring issue not seen for a threshold, e.g. 7-14 days depending on issue severity.
Returning issue	Dormant/resolving issue appears again 1-2 times in similar window.
Unresolved issue	Issue repeats and has no positive replacement/outcome after the event window.
Replacement detected	Positive Path action, journal, mood improvement, or user feedback occurs within 1-3 hours after issue.
Positive shift	Issue frequency drops while replacement behavior increases in same vulnerable window.
Needs care	High-care issue repeats, returns, or stays unresolved beyond threshold.

Positive replacement examples

- User used to drink at night, but now reads books at night.
- User used to log sadness, but now logs happy, blessed, calm, or grateful in the same window.
- User used to isolate, but now messages a friend or journals.
- User used to procrastinate, but now starts a 10-minute Path action.
- User used to experience anger after work, but now takes a walk or calm reset.

Pattern summary examples

Example 1 - resolving

Night sadness appeared 4 times last week. This week it appeared once, and reading at night appeared 3 times. Current read: positive replacement is forming.

Example 2 - returning

Evening alcohol craving was quiet for 12 days, then appeared twice this week around 8pm. Current read: returning pattern; protect the window early.

15. North Star integration

Brain 2.0 should not sit alone as a chart. It should shape North Star's daily language, notifications, goal suggestions, and follow-up behavior.

Morning letter inputs

- Today's likely vulnerable windows.
- Returning issue warnings.
- What helped before.
- One protective replacement action.
- Daily goals suggested from Aim, history, Path, and Life Map.
- Safety wording when high-care issues are involved.

Morning letter example

Morning letter

Today, protect the evening window. Loneliness has shown up there twice this week, and journaling helped once. The goal is not a perfect day - it is catching the loop early. I suggest three small goals today: 25 minutes coding, 10 minutes walking before 8pm, and a 3-line night note.

Evening letter inputs

- Issues logged today.
- Mahout interventions today.
- User responses and replacements.
- Positive effects and unresolved items.
- Whether a vulnerable window was protected, interrupted, or repeated.

Evening letter example

Evening letter

At 8pm the old craving window showed up again, but you interrupted it with a walk. That is not small. This is the loop changing. Tomorrow, we protect that window a little earlier.

North Star chat behavior

- Use Life Map evidence without sounding creepy.
- Say 'I am seeing a pattern' or 'your Life Map shows' when evidence is strong.
- Ask one question if confidence is soft.
- Offer direct action when readiness is high.
- Avoid endless clarification before giving value.

16. Key user flows

Flow A: user logs an issue

- 1 User logs 'loneliness' at night or writes it in chat/journal.
- 2 Brain 2.0 creates IssueEvent and candidate issue atom if needed.
- 3 Life Map updates the night cell.
- 4 If the issue matches a vulnerable window, North Star offers support or records the event for letter context.
- 5 Care Trail begins.

Flow B: Mahout starts a conversation

- 1 Pattern engine detects a high-confidence vulnerable window or returning issue.
- 2 Notification governance allows a conversation starter.
- 3 Mahout sends a gentle starter: 'This is usually a vulnerable window. How are you doing right now?'
- 4 User opens North Star. Live response may use credits or paid entitlement.
- 5 Outcome is tracked: opened, ignored, journaled, Path action, improved mood, issue returned.

Flow C: positive replacement forms

- 1 A negative issue used to happen at night.
- 2 User starts reading books at night or logs blessed/happy/calm.
- 3 Pattern engine detects issue frequency down and replacement frequency up in same window.
- 4 Life Map shows 'positive shift' or 'replacement forming'.
- 5 North Star reinforces it: 'This is becoming your new night pattern.'

Flow D: morning daily goals

- 1 North Star builds a daily plan from Aim, previous daily goals, Path history, Life Map, and current Brain.
- 2 User sees suggested goals/actions inside North Star.
- 3 User edits/deletes/adds/reorders.
- 4 User saves. Existing Today Goals/Path/Aim structures receive the saved plan through existing use cases.
- 5 Evening letter reviews what happened.

Flow E: auto-linking

- 1 User creates a goal or Path action.
- 2 AI/model-policy checks obvious semantic links.
- 3 High-confidence links are applied with undo.
- 4 Medium-confidence links ask one question.

- 5 Rejected links become negative feedback so Mahout avoids repeating them.

17. Safety, trust, and high-care boundaries

Brain 2.0 can handle real issues like alcohol cravings, sadness, loneliness, anger, low energy, and unhealthy loops, but it must avoid clinical claims. The app is support and pattern-awareness, not treatment.

High-care issue categories

- Alcohol craving or substance-related cravings.
- Self-harm or crisis signals.
- Severe depression or hopelessness language.
- Eating disorder-like behavior.
- Violence or danger signals.
- Trauma-sensitive topics.
- Anything the user marks sensitive.

Required safeguards

Safeguard	Behavior
Sensitive labels	Flag high-care issues so copy and prompts become careful.
Safety copy	Show: 'Mahout can help you notice patterns and choose safer next steps. This is not medical care.'
Crisis routing	If crisis/self-harm appears, use safe support flow and real-world help direction.
No diagnosis	Do not label user as addicted, depressed, or disordered.
No cure/treatment claims	Use 'seemed to help' and 'interrupt the loop'.
User controls	Hide, delete, exclude from letters, disable sparks/conversation starters.
Evidence transparency	Show why Mahout thinks something is a pattern.
No shame	Returning issue is not failure. It is an early catch signal.

High-care message example
This is a high-care pattern. Mahout can help you notice it early and choose safer next steps, but this is not medical care. If this feels hard to manage alone, reaching out to real-world support matters.

18. Practical build plan

Brain 2.0 should be built in surgical phases. Each phase must inspect the current repo first, detect what already exists, avoid duplicate systems, and implement only the smallest missing piece.

Phase	Goal	Done when
0. Repo audit	Find existing Brain atoms, memories, patterns, trends, monetization policy, letters, sparks, Path/Aim links.	We know exact seams and avoid duplicates.
1. Brain 2.0 scope constants	Add internal vocabulary, policy keys, and safe enums if missing.	No UI yet; terms are stable.
2. Memory Atoms 2.0	Add issue/trigger/replacement/outcome/time/safety/evidence atom support.	Old memory UI still works; new atoms can be stored.
3. Issue Events	Store explicit issue events with time, intensity, source, sensitivity, and evidence.	Events can be queried by day/week.
4. Pattern Engine	Compute Life Map cells, vulnerable windows, recurring/returning/resolving states.	Tests prove deterministic states.
5. Care Trail	Build structured issue -> intervention -> response -> outcome timeline.	Issue detail can show what Mahout did and what happened after.
6. Life Map UI	Premium visible 7-day map with cells, chips, bottom sheet, filters, empty state.	Paid user can see it; free sees lock/preview.
7. North Star letters	Inject high-confidence Life Map context into morning/evening letters.	Letters reference patterns safely and exclude hidden/deleted atoms.
8. Conversation starters	Use vulnerable windows and returning issues to start meaningful check-ins.	Governed, paid-only, non-spammy.
9. Action layer	Morning daily goals inside North Star with edit/save/link.	User can accept a useful plan without going to Aim/Path manually.
10. Auto-linking	AI-assisted goal/action linking with high-confidence auto-link and undo.	Receipts flow with less user friction.
11. Path guide	Add visual guide for time actions/check actions without changing Path logic.	New users understand Path quickly.
12. Trust controls	Hide/delete/exclude/safety footer/why-this evidence.	User controls memory and AI context.

MVP that is still amazing

- Memory Atom 2.0 support for issues, replacements, outcomes, and evidence.
- Issue Events stored and grouped into time blocks.
- Visible paid Life Map 7-day grid.
- Tap cell detail with evidence.
- Care Trail for each important issue.
- Returning Issue label and support copy.
- Morning/evening letter references for high-confidence patterns.

- Premium gate using existing monetization policy.
- Tests for vulnerable windows, returning issue, replacement detection, and hidden/deleted exclusion.

19. Cursor implementation rails for later

No Cursor prompts are included here yet. This section records the rules that every later prompt set should preserve.

Non-negotiables

- Kotlin + XML only. No Jetpack Compose.
- MVVM with Fragment/ViewBinding -> ViewModel -> UseCases -> Repository -> Room/DataStore as existing.
- Use Hilt and existing package conventions.
- Preserve monetization, credit ledger, entitlement resolver, Pro Light, Bond Pass, upgrade surfaces, and billing seams.
- Do not rebuild North Star chat, Brain, Path, Aim, or notifications broadly.
- Do not rename packages or move files unless absolutely required.
- Do not delete existing memory tables or Path/Aim behavior unless the repo proves unavoidable.
- Do not touch timer cadence, notification governance, billing math, credit spending rules, or model routing unless that phase explicitly requires it.
- All free/paid decisions go through the existing feature policy/resolver.
- All AI output that affects UI should be JSON-first or validated before rendering.

Prompt-set structure later

- 1 Prompt 1: set stage, inspect repo, identify existing seams, report exact plan before edits.
- 2 Prompts 2-8: one micro-slice each, with exact files likely touched.
- 3 Final prompt: strict verification, compile/test run, changed file list, deletions, risks, remaining gap.

Final principle

Brain 2.0 should feel like Mahout is remembering the user's real loops, showing them clearly, caring at the right time, and helping turn old patterns into better ones.

20. Sample copy bank

Vulnerable window

- This may be a vulnerable window. Want comfort, connection, or one tiny replacement action?
- This is usually the hour where this pattern gets louder. No shame - catching it early is the win.
- Before the window opens fully, let's protect it with one small move.

Returning issue

- This old pattern may be starting to return. We do not panic. We catch it early.
- This was quiet for a while, and now it has shown up twice again. Let's protect the next window.
- We have seen this loop before. Last time, the first useful move was small.

Positive replacement

- This is becoming a new pattern: reading is showing up where sadness used to show up.
- The old night loop did not disappear by magic. You replaced part of it.
- Blessed and calm showed up in the same window where loneliness used to return. That matters.

Daily goals

- I made a small plan for today based on your Aim, yesterday, and your Life Map. You can edit anything before saving.
- Today does not need a huge plan. It needs three clean receipts.
- I protected the evening window in this plan because that is where the old loop has been returning.

Auto-linking

- I linked Running and Workout to Reduce weight so your Path receipts count automatically.
- This action looks like it supports your Mahout programming goal. I linked it - undo anytime.
- I found a few actions that may feed this goal. Want me to link them?